

LAB 3: Problem Statement: Hostel Maintenance & Issue Ticketing System

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Problem Number: 7

GitHub link: <https://github.com/PSP-chillin>

System: Hostel Maintenance & Issue Ticketing System

The Hostel Maintenance & Issue Ticketing System is a hostel facility management platform that allows students and hostel administrators to report, track, and resolve maintenance issues. The system:

- Receives maintenance complaints/issues reported by hostel residents.
- Records and categorizes issues such as electrical, plumbing, cleaning, and room maintenance.
- Assigns reported issues to appropriate maintenance staff.
- Tracks the current status of maintenance tickets.
- Allows students to view the progress and status of their reported complaints.
- Provides hostel administrators with an overview of pending, ongoing, and resolved issues.
- Allows maintenance staff and hostel authorities to monitor and manage maintenance operations.

Key Functional Requirements

ID	Requirement
FR-001	The system shall allow hostel residents to report maintenance issues.
FR-002	The system shall categorize and record reported maintenance complaints.
FR-003	The system shall assign maintenance tickets to appropriate staff.
FR-004	The system shall allow residents to track the status of their complaints.

ID	Requirement
FR-005	The system shall allow hostel admins to manage & monitor the maintenance operations.

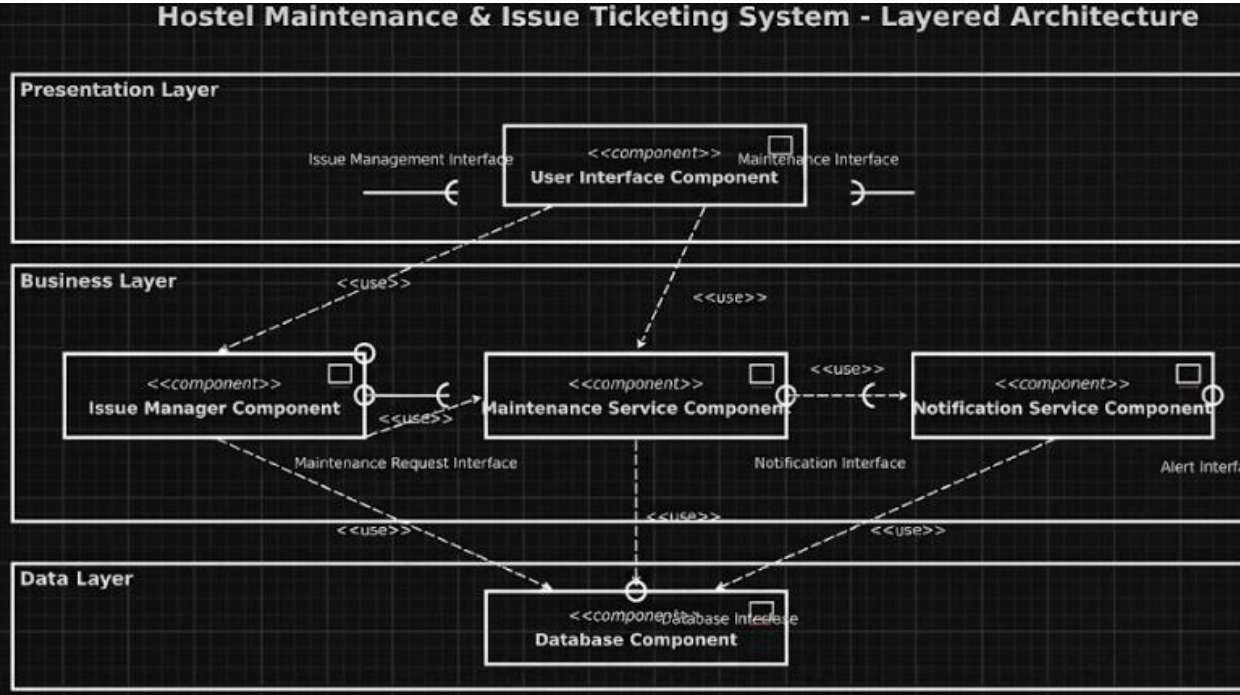
Key Non-Functional Requirements

Requirement	Importance
Performance	The system should process maintenance requests with low latency.
Scalability	The system should support an increasing number of hostel residents and maintenance requests.
Availability	The system should remain available even if one subsystem fails.
Security	Resident and maintenance data must be protected from unauthorized access.
Reliability	Maintenance ticket status and updates should be recorded accurately and consistently.

Architectural Style Analysis:

Architecture	Advantages for HMS	Disadvantages for HMS
Layered	Clear separation between presentation, business logic, and data; easy to understand, develop, and maintain.	Scaling individual functions such as ticket processing or notifications can be difficult.
Microservices	Individual services can scale independently; failures can be isolated; suitable for handling different hostel management functions.	More complex deployment and communication; network latency and data consistency must be managed.
Client-Server	Simple centralized architecture; easy data management and deployment.	Central server can become a bottleneck and single point of failure.

UML Diagram:



Selected Architecture — Layered Architecture

I chose Layered Architecture for the Hostel Maintenance & Issue Ticketing System.

Reason 1 – Separation of Responsibilities

The system separates the presentation, business logic, and data access into distinct layers. This makes the system easier to understand and allows changes in one layer to be made without significantly affecting the others.

Reason 2 – Maintainability

Maintenance operations such as issue reporting, ticket assignment, priority management, and status updates can be handled in the business logic layer. This makes the system easier to develop, test, debug, and maintain.

Security Advantage

Layered Architecture allows authentication and authorization to be handled within the application layers. Students, maintenance staff, and administrators can be given access only to the functions and data appropriate to their roles, protecting resident and maintenance information.

Performance Benefit

The layered structure provides efficient communication between the user interface, business logic, and database. Since the hostel system does not require large-scale real-time processing, Layered Architecture provides sufficient performance without the additional communication and deployment overhead of Microservices.

Conclusion

Layered Architecture provides the best balance of simplicity, maintainability, security, and performance for the Hostel Maintenance & Issue Ticketing System. Although it may be less flexible for independently scaling individual components, its clear separation of

presentation, business logic, and data access makes it highly suitable for a centralized hostel management system.